

TechReport Part1 BRS

Blinkit Sales Performance Analysis Project

Prepared by: Kemjira Nugboon Student ID 6688072



Project Summary

Business Domain	Quick Commerce
Company Name	Blinkit (Grofers India Pvt. Ltd.)
Program	Database Administration (DBA)
Project Number	BLINKIT-2026-001
Project Description	Design and management of Blinkit sales database to support BI & Data Warehouse systems, focusing on data integrity, performance optimization, inventory data handling, outlet operations, delivery records, and customer data management.

Document Control

Prepared by:	Kemjira Nugboon 6688072
Title:	Database Administrator (DBA)
Email:	kemjira.nug@student.mahidol.edu
Phone:	0918791107
Location:	Phutthamonthon Sai 4 Road ,Salaya, Phutthamonthon District, Nakhon Pathom
Version no:	0.1
Version date:	22/01/2026
Status:	Initial Draft
File/Doc no:	File number/document number

Version history

Version no.	Date	Changed by	Nature of amendment
0.1	22/01/2026	Kemjira	Initiated Part 1 (BRS): Defined project scope and initial business requirements.
0.2	12/02/2026	Kemjira	Commenced Part 2 (Database Design): Developed conceptual ERD and identified entities
0.3	25/02/2026	Kemjira	Finalized Part 2 Database Server Specification
0.4	31/03/2026	Kemjira	Developed Part 3 Maintenance
0.5	05/04/2026	Kemjira	Integrated Part 1 & Part 2: Updated system flowcharts and UML diagrams to ensure operational consistency.
0.6	15/04/2026	Kemjira	Optimized Part 3 Performance Tuning and refined the integrated report structure across all parts.
1.0	16/04/2026	Kemjira	Final proofreading and formatting for submission.

Executive summary

This project analyses the Blinkit sales dataset to support business decision-making in a quick-commerce environment. Blinkit operates as India's leading instant-delivery grocery platform, fulfilling orders within minutes. The report focuses on order processing, inventory management, delivery operations, and customer satisfaction. The outcome includes business requirements, process analysis, and data-driven insights to improve operational efficiency and database system design. Four core business requirements (RQ01–RQ04) have been identified, spanning order management, inventory tracking, delivery performance monitoring, and customer feedback management.

1. Purpose of this document

The purpose of this document is to define and analyse business requirements based on the Blinkit dataset to support data-driven decision-making and relational database system development.

2. Definitions

Terms	Meaning
Blinkit	Quick-commerce grocery delivery platform, delivering items within minutes.
BRS	Business Requirements Specification (this document).
DBA	Database Administrator, responsible for database design, maintenance, and performance.
BI	Business Intelligence, systems and tools for analysing business data.
DW	Data Warehouse, centralised repository for integrated, historical data.
OLTP	Online Transaction Processing, real-time transactional database system.
KPI	Key Performance Indicator, measurable value demonstrating business objective achievement.
SLA	Service Level Agreement, commitment for delivery time and service quality.
Quick-Commerce	A business model that focuses on delivering groceries and small consumer items in a very short time, often under 30 minutes
Net Usable Stock	The actual quantity of inventory available for sale, calculated by subtracting damaged goods from the total received stock

3. Project definition

3.1 Background

Blinkit operates as a quick-commerce platform providing rapid delivery of groceries and household items. The dataset represents sales transactions, inventory details, outlet types, and product categories. Analysing this data helps simulate real-world retail business challenges.

3.2 Objectives

- To analyse sales performance across product categories.
- To identify inventory patterns and track net usable stock.
- To monitor delivery efficiency and identify service-level deviations.
- To assess customer satisfaction trends using rating and feedback data.
- To define business requirements supporting operational databases
- To support management decision-making using data-driven insights.

3.3 Project performance measurement/success criteria/KPIs

- Achievement of 100% requirements coverage by mapping the four defined Business Processes from BP01 to BP04 directly to the physical database schema.
- Fulfillment of Database Normalization standards by ensuring all core tables meet Third Normal Form (3NF) requirements to effectively eliminate data redundancy.
- Maintenance of strict Data Integrity during the migration and mapping of the Kaggle dataset into the relational schema with zero critical data anomalies.
- Completion of all Documentation requirements including the delivery of comprehensive Technical Reports for BRS, logical design, and DBA maintenance tasks.
- Optimization of system performance through strategic indexing and query tuning to ensure efficient data retrieval for sales, inventory, and delivery reporting, validated by quantifiable performance gains.

3.4 Dataset Description

The dataset used in this project is the Blinkit dataset obtained from Kaggle. The dataset consists of multiple CSV files representing different aspects of business operations, including orders, products, inventory, delivery, and customer feedback.

To comply with the project requirement, only five relevant files are selected for database design and analysis as follows:

- blinkit_orders.csv
- blinkit_order_items.csv
- blinkit_products.csv
- blinkit_inventory.csv
- blinkit_customer_feedback.csv

These selected files support the core business processes including order management, inventory tracking, and customer feedback analysis. All files are in CSV format and are suitable for DBMS import.

The remaining files are not used in this project due to scope limitation and relevance:

- blinkit_customers.csv (excluded as customer details are not the main focus)
- blinkit_delivery_performance.csv (excluded to limit system scope)
- blinkit_inventoryNew.csv (excluded due to duplication with inventory data)
- blinkit_marketing_performance.csv (excluded as marketing analysis is outside the project scope)

4. Project scope

4.1 In scope

- Tracking order processing across various product categories.
- Managing stock levels, including the recording of received quantities and calculation of damaged goods to determine net usable stock.
- Monitoring delivery performance by recording driver assignments and comparing promised versus actual delivery timestamps.
- Analyzing customer satisfaction through a structured rating and feedback validation process.
- Development of the Entity-Relationship Diagram (ERD) and Logical Schema (3NF) to support the core business functions.
- Specification of hardware/software requirements, database maintenance planning, and performance tuning strategies.

4.2 Out of scope

- Real-time synchronization with Blinkit's production environment or live APIs.
- Management of customer login credentials, passwords, or advanced identity security protocols.
- Direct integration with external delivery partner platforms, as driver data is treated strictly as manual textual input.
- Financial accounting, payroll systems, and human resource management.
- Creation of mobile applications or consumer-facing web interfaces.
- Deployment of Machine Learning models for demand forecasting.

4.3 Constraints

- The analysis and system design are limited to a static historical data snapshot, as real-time synchronization with live operational databases is not included in this initial phase.
- Hardware and infrastructure planning must adhere to strict initial budget allocations, focusing on cost-effective, single-server deployment over immediate enterprise-level scaling.
- The system architecture is currently constrained to centralized processing, deferring distributed database or cloud-cluster expansion to future development cycles.
- All implementation and testing phases must be executed and finalized within the strictly defined project schedule to meet organizational rollout objectives.

4.4 Assumptions

- The historical sales and transaction records provided for system design are assumed to accurately reflect actual daily operational volumes and business behaviors.
- Customer satisfaction ratings uniformly utilize a standardized 1–5 numerical scale across all service channels.
- When a customer adds an item to their digital cart via the application, the backend system immediately records the unique Item_Identifier into the Order_Detail database table to guarantee accurate price retrieval and execute flawless billing calculations for the transaction.
- System calculations for delivery SLAs and net usable inventory are entirely dependent on accurate timestamp logs and stock entry inputs from operational staff.

5. Stakeholder impacts

The following internal and external stakeholders are identified as having an interest in or being impacted by this project:

Internal

Stakeholder	Impact/Interest in the project
Sales and Order Management Team	Requires accurate order tracking and sales trend reports to optimize daily operations and maximize revenue.
Inventory and Warehouse Team	Relies heavily on real-time inventory monitoring to trigger replenishment cycles and maintain an accurate count of net usable stock.
Customer Service Team	Manages customer feedback, resolves negative ratings, and utilizes system data to handle service disputes efficiently.
Database Administrator (DBA)	Responsible for designing, maintaining, and tuning the core database infrastructure to ensure system stability and analytical performance.
Senior Management / Executive Team	Consumes high-level database reports and executive summaries for strategic planning and resource allocation.

External

Stakeholder	Impact/Interest in the project
Customers	Directly impacted by the system's ability to ensure product availability, maintain order accuracy, and fulfill rapid delivery SLAs.
Delivery Partners	Their operational performance (pickup and delivery timestamps) is systematically recorded to evaluate SLA compliance and service quality.
Suppliers / Vendors	Affected by inventory replenishment thresholds, as accurate stock data drives timely purchase orders and supplier relationships.
Regulatory Bodies	Requires the business to ensure that all data governance and consumer information handling strictly comply with applicable privacy laws and digital commerce regulations.

6. Business processes: Overview concept

The Blinkit Sales Dataset contains transactional and operational data from a quick-commerce grocery business, including orders, inventory , delivery performance, and customer feedback. The dataset represents daily retail operations and supports the following key business processes:

Process no.	Business process name	Impact area/division/sector	Reference
BP01	Order Process	Order Management / Sales	RQ01
BP02	Inventory Tracking Process	Inventory Control / Warehouse	RQ02
BP03	Delivery Performance Process	Delivery Operations	RQ03
BP04	Customer Feedback Process	Customer Insights	RQ04

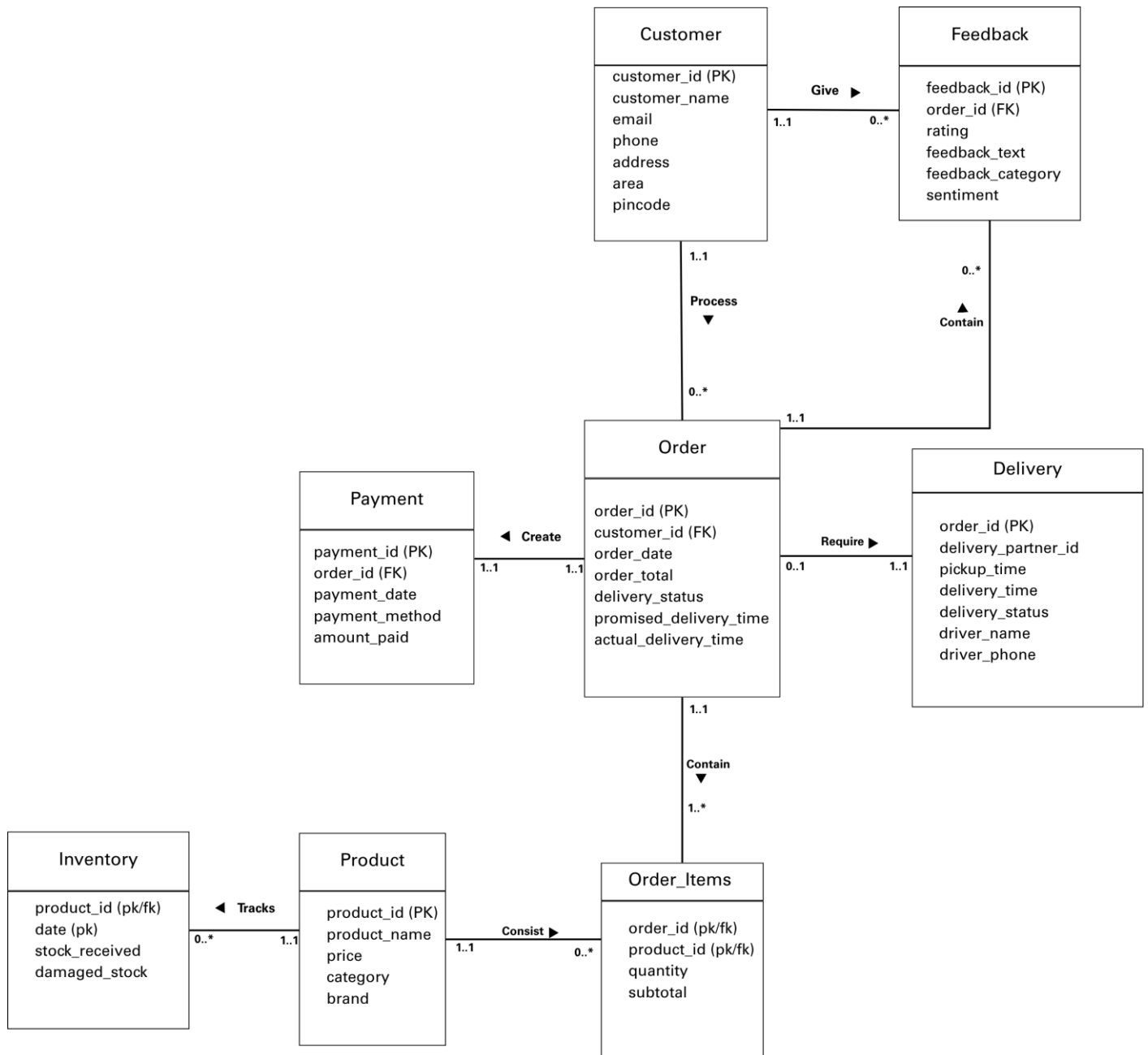
BP01: Order Process manages the customer's journey of placing an order, including the validation of order details, calculation of the total amount, and secure payment processing before transferring the data to the delivery stage.

BP02: Inventory Tracking Process handles the reception of incoming stock, inspects for damaged goods to calculate net usable stock, updates overall inventory levels, and triggers replenishment alerts when stock falls below the minimum threshold.

BP03: Delivery Performance Process oversees the assignment of delivery partners, tracks the pickup and delivery status in real-time, handles any delays or failures, and records completion times to evaluate service efficiency.

BP04: Customer Feedback Process manages the collection of post-delivery reviews, requests corrections for any invalid inputs, and securely stores the validated feedback data for future service improvement analysis.

6.1 Entity Relationship Diagram (ERD)



The Entity Relationship Diagram (ERD) presents the overall structure of the database system and shows the relationships among key entities such as Orders, Order_Items, Products, Delivery, Inventory, and Customer Feedback. The diagram reflects the core business processes of the system and defines relationships including one-to-many and one-to-one associations. These relationships help maintain data consistency, reduce redundancy, and support efficient data management within the database system.

7. Business requirements

The business requirements in this section are derived from the analysis of the Blinkit sales and the associated business processes. These requirements are designed to support key business functions, including order management, inventory tracking, delivery operations, and customer feedback evaluation. Each requirement is clearly defined, business-oriented, and can be validated through measurable analytical reports and performance insights.

RQ01: Manage Order Data	
Reference	RQ01
Business Requirement Name	Manage Order Data
Benefit	Mandatory
Status	In Scope
Cross Reference to Business Process	BP01 – Order Process
Purpose	To ensure accurate recording and management of order data for operational use and performance analysis.
As-Is Business Requirement	Order data is stored in the system but lacks consistency and validation for reliable usage.
To-Be Business Requirement	The system will store and manage complete and accurate order data, enabling reliable reporting and analysis of order activities.
Business Rules	• Each order must have a unique order_id. • Order total must be non-negative. • Order date must be recorded. • Each order must be linked to valid product data.
Issues	Data inconsistency or missing fields may affect data quality and reporting accuracy.

RQ02: Track Inventory Movements	
Reference	RQ02
Business Requirement Name	Track Inventory Movements
Benefit	Mandatory
Status	In Scope
Cross Reference to Business Process	BP02 – Inventory Tracking Process
Purpose	To monitor stock levels, stock movement, and damaged inventory to ensure product availability.
As-Is Business Requirement	Inventory data is available but not effectively tracked for stock movement and loss analysis.
To-Be Business Requirement	Inventory data will be tracked and analysed, including stock received and damaged stock, to maintain accurate inventory records and support operational decisions.
Business Rules	• Each inventory record must include product_id and date. • stock_received must be greater than or equal to 0. • damaged_stock must be greater than or equal to 0. • damaged_stock must not exceed available stock.
Issues	No issues

RQ03: Monitor Delivery Performance	
Reference	RQ03
Business Requirement Name	Monitor Delivery Performance
Benefit	Mandatory
Status	In Scope
Cross Reference to Business Process	BP03 – Delivery Performance Process
Purpose	To evaluate delivery efficiency and identify delays in order fulfilment.

As-Is Business Requirement	Delivery data is collected but not analysed to identify performance issues.
To-Be Business Requirement	Delivery data will be analysed to track delivery status, measure delivery time, and identify delays for operational improvement.
Business Rules	• Each delivery must be linked to a valid order_id. • Delivery status must be recorded. • Delivery time must be greater than 0.
Issues	No issues

RQ04: Analyse Customer Feedback	
Reference	RQ04
Business Requirement Name	Analyse Customer Feedback
Benefit	Optional
Status	In Scope
Cross Reference to Business Process	BP04 – Customer Feedback Process
Purpose	To evaluate customer satisfaction and identify service improvement opportunities.
As-Is Business Requirement	Customer feedback is collected but not analysed systematically.
To-Be Business Requirement	Customer feedback and ratings will be analysed to measure satisfaction levels and identify trends.
Business Rules	• Feedback must be linked to a valid order_id. • Rating must be within a valid range (1–5). • Each order can have at most one feedback record.
Issues	Limited feedback data may reduce the reliability of customer satisfaction insights.

8. Business processes

This section presents a detailed description of the business processes associated with the defined business requirements. Each process is analysed in terms of its inputs, activities, and outputs to ensure clear definition and logical sequencing. The processes support key operational and analytical functions, including order management, inventory tracking, delivery operations, and customer feedback evaluation. These processes reflect real-world operations in a quick-commerce environment and ensure alignment between business activities and data requirements.

8.1 Order Process

8.1.1 Description

This process manages customer orders from the point of order placement through validation and confirmation within the system. It ensures that all order data is accurately captured, recorded, and prepared for downstream operations such as inventory allocation and delivery fulfilment. The process plays a critical role in supporting sales operations and maintaining data consistency for reporting and analysis.

From business requirement	RQ01 – Manage Order Data
---------------------------	--------------------------

8.1.2 Event sequence

Preconditions	
- Customers are able to access the ordering platform - Product information is available and valid - The system is operational and able to record transactions	

Typical course of events	
Step	Action
Step 1	Customer places an order
Step 2	System generates unique order ID
Step 3	System records order date and store information
Step 4	System validates order information
Step 5	System calculates total order amount
Step 6	Customer selects payment method and processes payment
Step 7	System confirms successful payment
Step 8	System proceeds to Delivery Performance Process (8.3)

Alternate course of events	
Step	Action
Alternate A at step 4	Invalid order information
Step A1	System identifies invalid order details (e.g., negative total or missing products)
Step A2	System notifies the customer and requests correction
Step A3	Customer modifies the order, and process resumes from the validation step

Alternate B at step 6	Payment failed
Step B1	System detects payment failure
Step B2	System prompts customer to select another payment method
Step B3	Process resumes from step 6

8.1.3 Graphical representation

This diagram illustrates the end-to-end order process, including order placement, validation, confirmation, and transition to fulfilment.

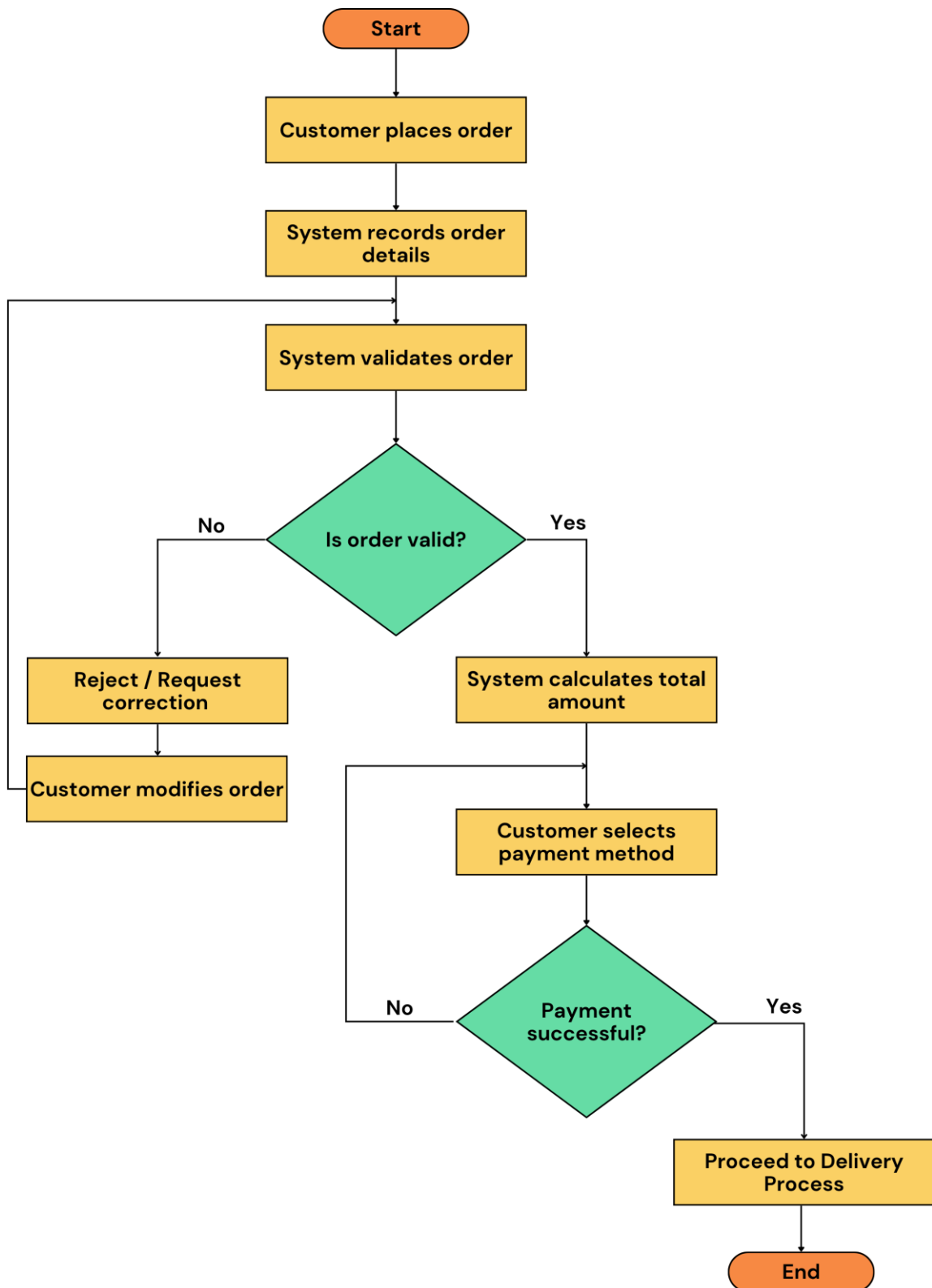


Figure 8.1: Flowchart for Order Process

8.2 Inventory Tracking Process

8.2.1 Description

This process manages and tracks inventory movements within the system, including stock received, stock availability, and damaged stock. It ensures that inventory records remain accurate and up to date, enabling efficient order fulfilment and operational decision-making. The process supports stock monitoring and helps prevent stock shortages or inconsistencies.

From business requirement	RQ02 – Track Inventory Movements
----------------------------------	----------------------------------

8.2.2 Event sequence

Preconditions	
- Product records exist in the system - Inventory tracking is enabled for all products - Warehouse or storage operations are active	

Typical course of events	
Step	Action
Step 1	System receives stock entry data including the Product ID, date, and received quantity
Step 2	System prompts for an inspection of the incoming stock condition
Step 3	System records the identified damaged quantity from the inspection
Step 4	System calculates the net usable stock (received quantity minus damaged quantity)
Step 5	System updates the inventory records for the specific product
Step 6	System checks if the total stock level is below the minimum threshold

Alternate course of events	
Step	Action
Alternate A at step 3	
No damaged stock found	
Step A1	User enters 0 for the damaged quantity
Step A2	System processes the total received quantity as usable stock
Step A3	Process resumes from step 4

Alternate B at step 6	
Stock level is below minimum threshold	
Step B1	System detects that the total inventory is below the set threshold.

Step B2	System triggers a replenishment alert to the administrator
Step B3	Process ends

8.2.3 Graphical representation

This diagram illustrates the inventory tracking process, including stock updates, stock movement recording, and damage handling.

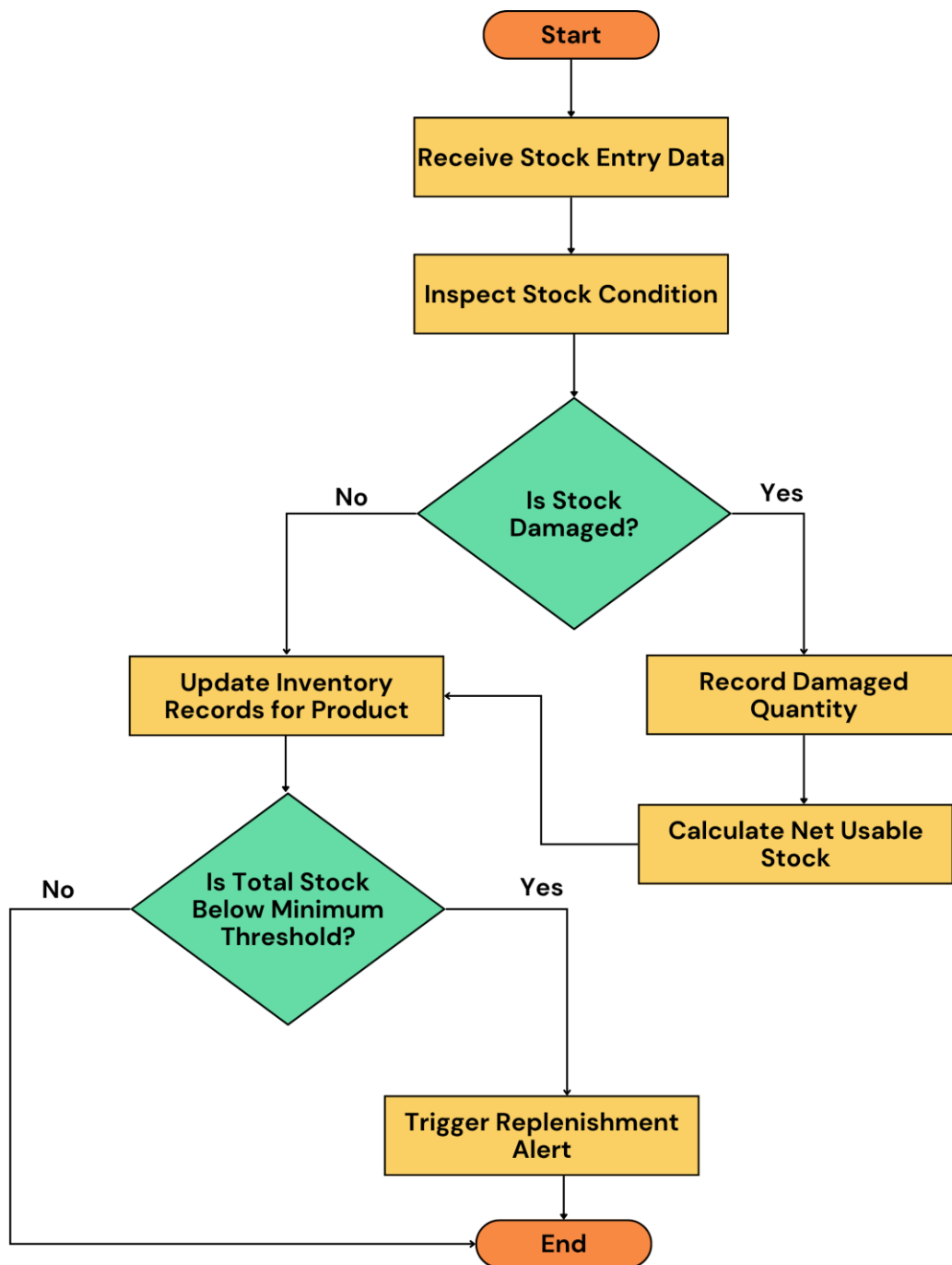


Figure 8.2: Flowchart for Inventory Tracking Process

8.3 Delivery Performance Process

8.3.1 Description

This process manages the delivery of orders from the outlet to the customer and tracks delivery performance. It ensures that delivery operations are executed efficiently and that delivery data is recorded for performance monitoring. The process supports the evaluation of delivery time, delays, and service efficiency.

From business requirement	RQ03 – Monitor Delivery Performance
----------------------------------	-------------------------------------

8.3.2 Event sequence

Preconditions	
- Order has been confirmed and is ready for delivery - Delivery resources (e.g., delivery partners) are available - Delivery tracking system is operational	

Typical course of events	
Step	Action
Step 1	System receives confirmed order for delivery
Step 2	System assigns delivery partner
Step 3	Delivery partner picks up the order
Step 4	Delivery partner delivers the order to the customer
Step 5	System records the delivery time
Step 6	System updates the final delivery status

Alternate course of events	
Step	Action
Alternate A at step 2	
Partner not available	
Step A1	System detects no partner is currently available
Step A2	System waits for 5 minutes and retries the assignment
Step A3	If maximum retries are reached, the system cancels the order and the process ends
Step A4	If a partner is found within the retry limit, the process resumes from step 3

Alternate B at step 3	Pickup delayed
------------------------------	----------------

Step B1	System detects a delay in order pickup.
Step B2	System updates the delivery status to reflect the delay.
Step B3	Delivery continues, and the process resumes from step 4.

Alternate C at step 4	Delivery unsuccessful
Step C1	System detects that the delivery was unsuccessful.
Step C2	System reschedules the delivery.
Step C3	System loops back to step 2 to reassign a delivery partner.

8.3.3 Graphical representation

This diagram illustrates the delivery process, including assignment, transportation, delivery completion, and performance tracking.

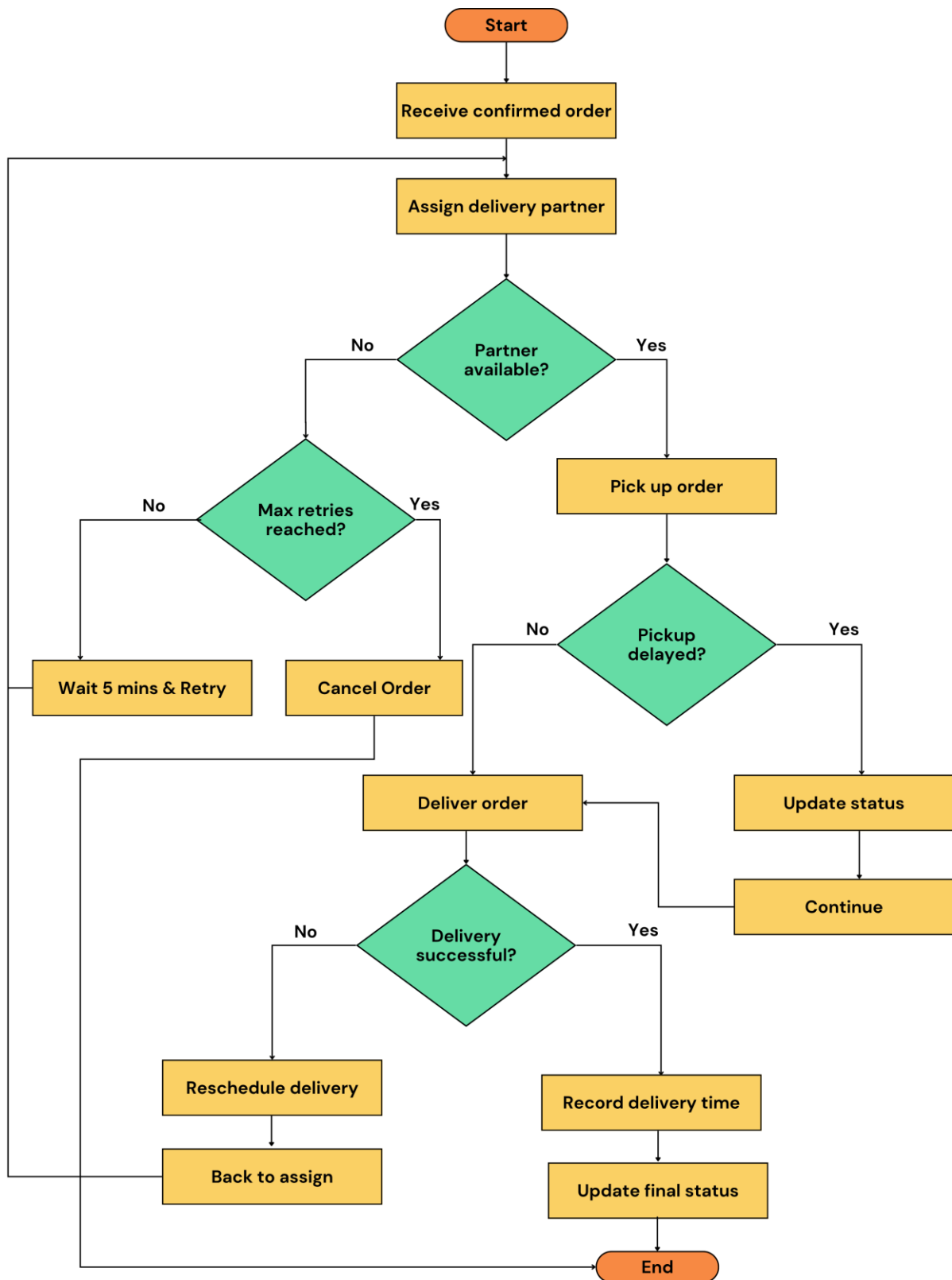


Figure 8.3: Flowchart for Delivery Performance Process

8.4 Customer Feedback Process

8.4.1 Description

This process collects customer feedback after order completion to evaluate customer satisfaction and service quality. It ensures that feedback data is captured, stored, and made available for analysis. The process supports continuous improvement of service performance and customer experience.

From business requirement	RQ04-Analyse Customer Feedback
----------------------------------	--------------------------------

8.4.2 Event sequence

Preconditions	
- Order has been successfully delivered - Customer has access to provide feedback - Feedback system is available	

Typical course of events	
Step	Action
Step 1	System identifies completed delivery
Step 2	Customer receives order
Step 3	System sends feedback request
Step 4	Customer opens feedback interface
Step 5	Customer submits rating
Step 6	Customer provides optional comments
Step 7	System validates feedback input
Step 8	System records feedback data
Step 9	System stores feedback for analysis

Alternate course of events	
Step	Action
Alternate A at step 5	
Step A1	Customer does not respond
Step A2	System marks feedback as missing
Step A3	Process ends

Alternate B at step 7		Invalid feedback input
Step B1	System detects invalid input	
Step B2	System requests correction from the customer	
Step B3	Customer modifies the feedback, and the process resumes from step 7	

8.4.3 Graphical representation

This diagram illustrates the customer feedback process, including feedback collection, validation, and storage.

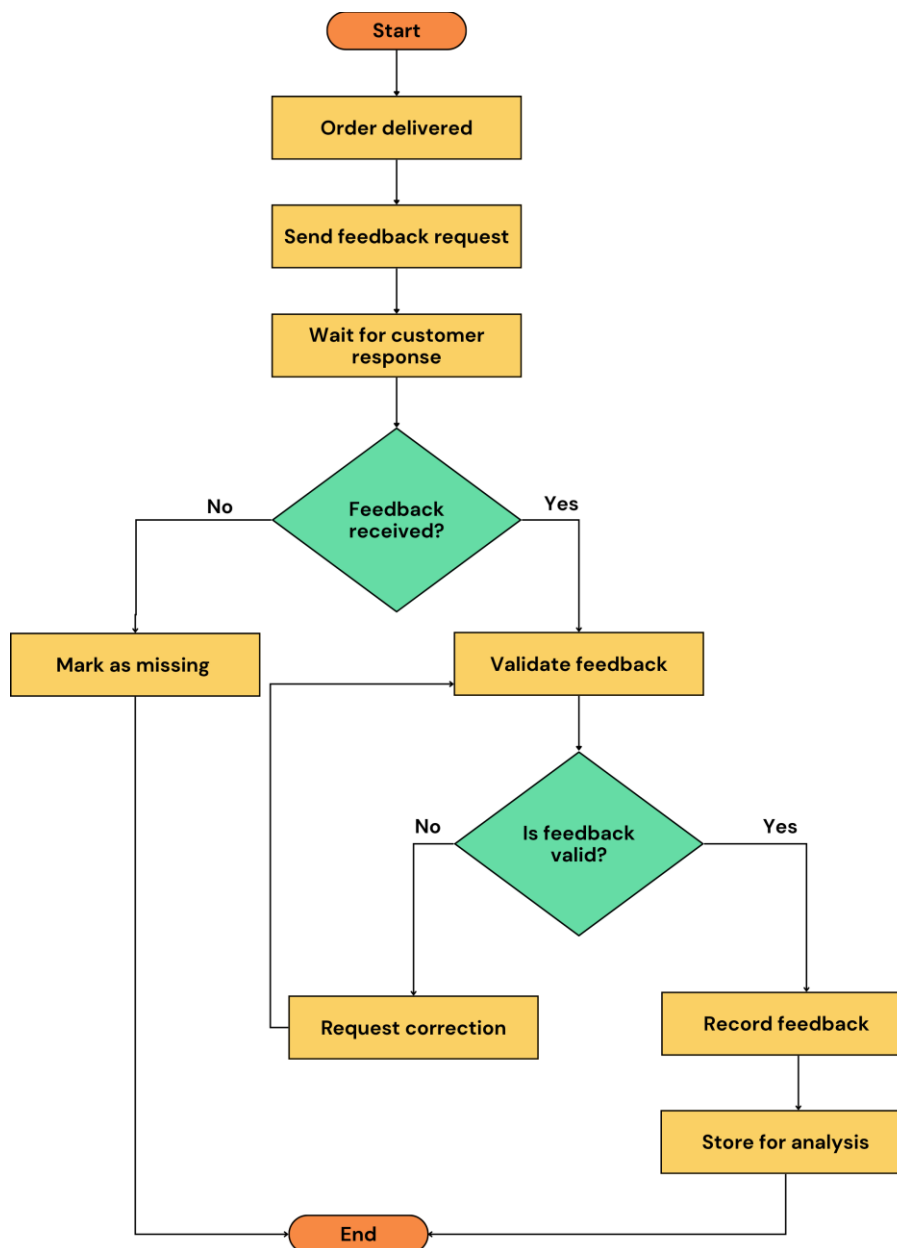
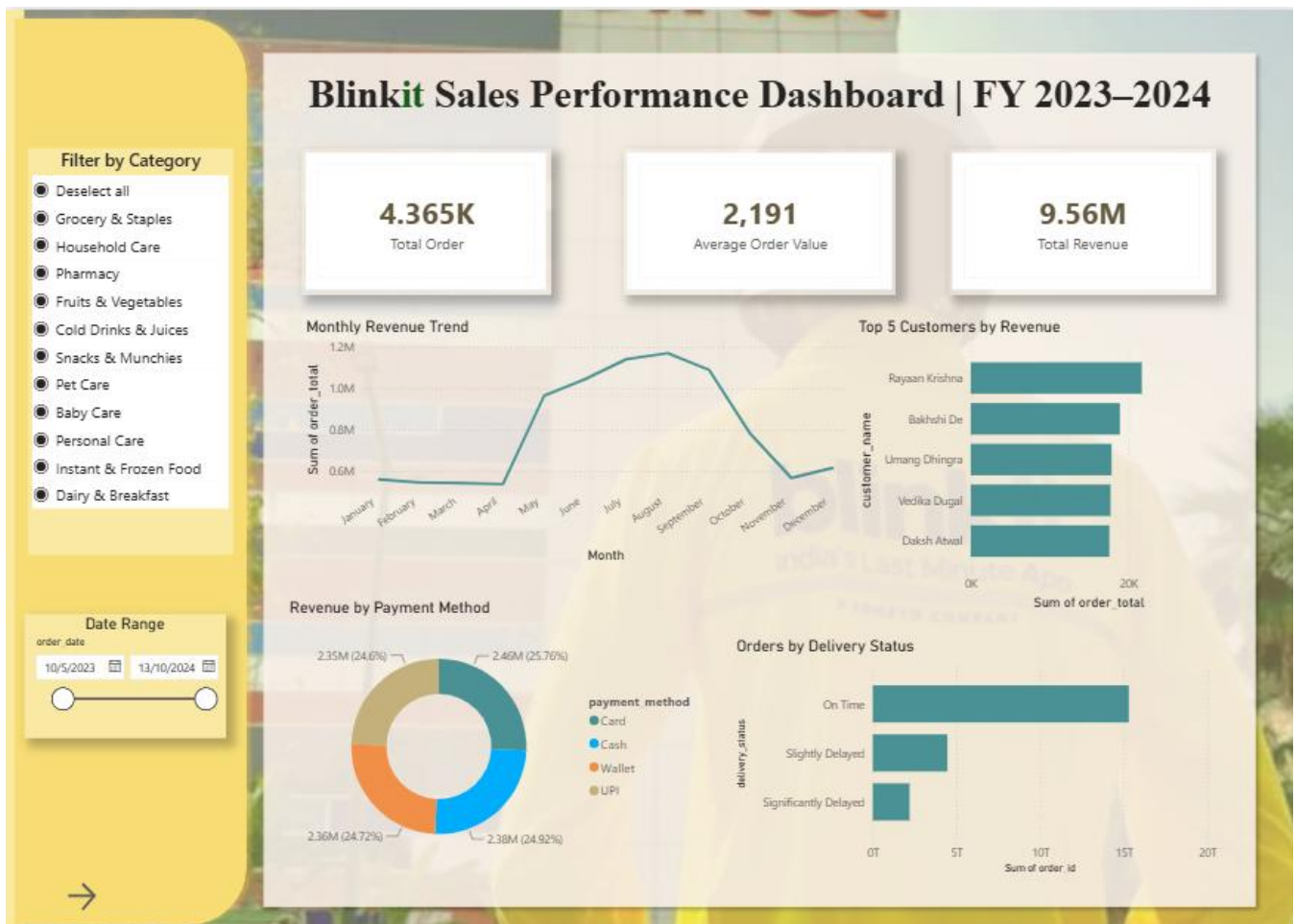


Figure 8.4: Flowchart for Customer Feedback Process

Appendix A: Power BI Sales Performance Dashboard



- Developed an interactive Power BI dashboard analyzing 4,365 orders and 9.56M in revenue across a 14-month period (2023–2024)
- Built KPI tracking for revenue trends, payment methods, and delivery performance, identifying mid-year (Jul–Aug) as peak revenue period
- Designed category and product-level breakdowns with customer segmentation, surfacing top 5 revenue-driving customers to support targeted business decisions

TechReport Part2 Database Server Specification

Blinkit Sales Performance Analysis Project

Prepared by: Kemjira Nugboon Student ID 6688072

1. Purpose of this document

The purpose of this document is to provide the technical specifications for the database server and environment required to host the Blinkit Sales Performance Analysis system. This includes hardware, software, and general constraints necessary to ensure data integrity, system performance, and reliable database administration for managing order, inventory, and delivery operations.

2. Definitions

Terms, abbreviations and acronyms	Meaning
ECC Memory	Error-Correcting Code memory, a type of RAM that detects and fixes internal data corruption.
IOPS	Input/Output Operations Per Second, a performance measurement used to characterize computer storage devices.
NVMe SSD	Non-Volatile Memory Express, a high-speed transfer protocol for SSDs that significantly reduces latency.
RAID 1	Redundant Array of Independent Disks (Mirroring), a configuration that duplicates data on two drives for fault tolerance.
L3 Cache	A specialized memory bank in the CPU that stores frequent data to speed up processing.
UPS	Uninterruptible Power Supply, a device that provides emergency power when the main power source fails.
LTS	Long-Term Support, a version of software (like Ubuntu) that is maintained for an extended period.
Indexing	A database optimization technique that speeds up data retrieval operations on a table.

3. General Description

3.1 System Overview

The Blinkit Sales Performance Analysis system is a robust relational database solution engineered to store, manage, and analyze complex quick-commerce data. The system focuses on integrating sales transactions, product inventory tracking, customer feedback, and delivery performance metrics. It is designed to be deployed on a dedicated database server, providing a centralized data repository accessible by authorized DBA staff and analytical query tools to support data-driven business decisions.

3.2 System Functions

- Store and manage comprehensive sale transaction records including order headers, itemized details, and payment methods.
- Maintain a centralized product master list containing item categories and specifications such as fat content and pricing attributes.

- Track real-time inventory levels including the recording of stock inflows and the calculation of damaged goods to determine net usable stock.
- Record and query customer satisfaction ratings and feedback to monitor service quality and identify areas for improvement.
- Monitor delivery operations by logging driver assignments and analyzing the variance between promised and actual delivery timestamps.
- Facilitate complex aggregation and relational join queries to generate operational reports and management executive summaries.
- Enforce strict data integrity and referential consistency through the implementation of primary keys, foreign keys, and predefined business rule constraints.

4. General Constraints

4.1 Hardware Constraints

- The core Database Management System must be MySQL 8.0 or later to ensure access to modern relational features and academic accessibility.
- The primary operating system must be a 64-bit Linux distribution such as Ubuntu 22.04 LTS to provide a stable and secure environment for database hosting.
- All development and data manipulation activities must utilize ANSI SQL standards to ensure cross-platform portability and industry-standard compliance.
- The system must utilize automated backup utilities capable of generating scheduled SQL dump exports to support disaster recovery and data archival policies.

4.2 Software Constraints

- The DBMS must be MySQL 8.0 or later (open-source, academically accessible).
- The operating system must be a 64-bit Linux distribution (Ubuntu 22.04 LTS recommended).
- All SQL queries must comply with ANSI SQL standards for portability.
- Backup software must support scheduled automated exports in SQL dump format.

5. Core Hardware Specification

5.1 CPU

The database server must support high-speed transactional processing and analytical queries for Blinkit sales and inventory data. The CPU should be capable of handling concurrent users and multiple database operations such as inserts, updates, joins, filtering, and reporting queries.

CPU Specification

Category	Specification	Notes
CPU Model	Intel Xeon / AMD EPYC (Server Grade) or Intel Core i7/i9 (Latest Gen)	Server-grade preferred for reliability and ECC memory support.
Cores / Threads	Minimum 8 Cores / 16 Threads	Supports concurrent OLTP and reporting workloads.
Clock Speed	3.0 GHz base or higher	Ensures fast single-threaded query execution.
L3 Cache	Minimum 16 MB	Reduces memory latency for frequent queries.

5.2 RAM / Memory Specification

The database system requires sufficient memory to store frequently accessed data in cache and reduce disk read operations. Since the Blinkit dataset contains multiple attributes related to deliveries, and order, memory performance is critical for query optimization.

RAM Specification

Category	Specification	Notes
Minimum RAM	16 GB	Suitable for development and testing environments.
Recommended RAM	32 GB	Required for concurrent analytical queries and BI workloads.
Memory Type	DDR4 / DDR5	DDR5 preferred for higher bandwidth.
Speed	3200 MHz or higher	Reduces query latency for in-memory operations.
ECC Support	Recommended	Error-Correcting Code memory prevents data corruption.

5.3 Storage Specification

Category	Specification	Notes
Hard disk type	NVMe SSD	High speed for database read/write operations.
Capacity	1 TB	Supports dataset growth and long-term historical records.
Speed	3000 MB/s+	Improves transaction processing and query response time.
Internal storage	NVMe SSD	Primary storage for active database files.
External storage	Cloud / External HDD	Used for daily backup and disaster recovery.
Throughput/IOPS	High IOPS	Suitable for OLTP (Online Transaction Processing) systems.
RAID Type	RAID 1	Provides fault tolerance and data reliability (Mirroring).

5.4 Overall Hardware Specification

Category	Specification	Notes
Role	Database Server	Supports OLTP (Online Transaction Processing) + reporting workloads.
CPU	8 Cores / 16 Threads	Capable of handling concurrent database queries efficiently.
RAM	32 GB DDR4/DDR5	Large memory capacity improves caching and query performance.
Boot Drive	256 GB SSD	Dedicated storage for Operating System and DBMS software.
Data Drive	1 TB NVMe SSD	High-speed storage for main database files and transaction logs.
Network	1 Gbps (min) / 10 Gbps (rec)	Supports multiple concurrent user connections and data transfer.
Power	UPS Support	Essential to prevent data loss or corruption during power outages.

6. Software Specification

6.1 Data Architecture and Data Model

Category	Specification	Notes
Data Model	Relational (SQL)	Tables with defined primary and foreign key relationships.
DBMS Family	MySQL 8.0+	Open-source, widely supported, ANSI SQL compliant.
Operating System	Ubuntu 22.04 LTS (64-bit)	Stable Linux environment for production database servers.
Query Language	SQL (Structured Query Language)	Standard language for data definition, manipulation, and control.
Backup Tool	mysqldump / MySQL Workbench	Automated scheduled SQL dump backups.
Monitoring Tool	MySQL Performance Schema + Slow Query Log	Built-in performance monitoring and slow query identification.

7. Functional Requirement: System Master List

Req.ID	Requirement Name	Requirement Description
REQ1-1	Manage Orders	The system must record and manage order and sales transaction data including itemized values and payment methods to support accurate sales reporting and operational tracking.
REQ2-1	Manage Inventory Levels	The system must store and maintain inventory and product-related information including damaged goods calculation to support stock monitoring and replenishment planning.
REQ3-1	Monitor Delivery Performance	The system must track delivery performance metrics such as driver assignments and delivery timestamps to identify delays and evaluate operational efficiency.
REQ4-1	Analyse Customer Feedback	The system must collect and analyze customer satisfaction ratings and feedback linked to specific orders to measure service quality and identify improvement areas.

8. Functional Requirement REQ1-1 Manage Orders

Process no.	Business process name	Impact area/division/sector
P01	Record Sales Transactions	Sales Operations / Order Management
P02	Validate Order Data	Database / Order Management
P03	Calculate Order Total	Sales Operations
P04	Store Order Record	Database / Order Management

9. Functional Requirement REQ2-1 Manage Inventory Levels

Process no.	Business process name	Impact area/division/sector
P01	Record Stock Received	Inventory / Warehouse
P02	Record Damaged Stock	Inventory / Warehouse
P03	Calculate Net Usable Stock	Inventory Control
P04	Update Inventory Records	Database / Inventory Management
P05	Monitor Stock Levels	Inventory / Warehouse

10. Functional Requirement REQ3-1 Monitor Delivery Performance

Process no.	Business process name	Impact area/division/sector
P01	Assign Delivery Partner	Operations / Logistics
P02	Track Delivery Status	Operations / Logistics
P03	Record Delivery Time	Database / Delivery Management
P04	Evaluate Delivery Performance	Operations / Analytics

11. Functional Requirement REQ4-1 Analyse Customer Feedback

Process no.	Business process name	Impact area/division/sector
P01	Collect Customer Feedback	Customer Service
P02	Validate Feedback Data	Database / Customer Insights
P03	Store Feedback Records	Database / Customer Insights
P04	Analyse Customer Satisfaction	Customer Insights / Analytics

12. Database Configuration and Performance Optimization

12.1 Database Configuration

Performance optimization in MySQL 8.0 focuses on efficient memory management and indexing strategies since the query cache feature has been removed in newer versions. The configuration emphasizes the allocation of the InnoDB buffer pool and management of concurrent connections to handle high-transaction workloads.

```
SHOW VARIABLES LIKE 'innodb_buffer_pool_size';
SHOW VARIABLES LIKE 'max_connections';
```

Variable_name	Value	Variable_name	Value
innodb_buffer_pool_size	134217728	max_connections	151

Figure 8.1: System variable configuration for memory and connection management

The current screenshot shows the default system value (128MB) during the development phase. However, for the production environment with 32GB RAM specified in Section 5.4, the recommended value is 24GB (25,769,803,776 bytes) to maximize memory-based data caching.

12.2 Query Performance Testing

The implementation of indexing demonstrates a significant improvement in database responsiveness. The execution time for a complex join query was reduced from 0.005686 seconds to 0.001982 seconds. This represents an efficiency gain of approximately 65% by reducing full table scans and optimizing the search path.

27	0.00568600	SELECT d.delivery_status, COUNT(*) FROM d...
28	0.02409200	CREATE INDEX idx_order_id ON order_items(o...
29	0.01056300	CREATE INDEX idx_delivery_order ON delivery...
30	0.00953900	CREATE INDEX idx_status ON delivery(deliver...
31	0.00198200	SELECT d.delivery_status, COUNT(*) FROM d...

Figure 12.2: Query performance comparison before and after indexing using profiling results

12.3 Index Implementation and Query Optimization

To improve database performance, indexing strategies are applied to frequently accessed columns such as `order_id`, `product_id`, and `delivery_status`. These indexes reduce full table scans and significantly enhance query execution speed.

Example of index creation:

```
CREATE INDEX idx_order_id ON order_items(order_id);
```

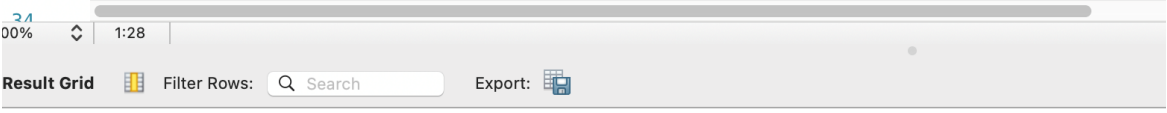
```
CREATE INDEX idx_delivery_order ON delivery(order_id);
```

```
CREATE INDEX idx_status ON delivery(delivery_status);
```

Query optimization is further analyzed using the EXPLAIN statement, which helps evaluate how MySQL executes queries.

Example :

```
31 • EXPLAIN SELECT * FROM Orders o
32 JOIN Delivery d ON o.order_id = d.order_id;
33
```



id	select_type	table	partitions	type	possible_keys	key	key_len	ref	rows	filtered	Extra
1	SIMPLE	o		ALL					1061	100.00	Using where
1	SIMPLE	d		ref	idx_delivery_order	idx_delivery_order	5	tech2.o.order_id	1	100.00	

The EXPLAIN output provides key insights including the type of query execution, index usage, number of rows scanned, and additional execution details.

- Before indexing, the query performs a full table scan (type: ALL), resulting in a higher number of rows being processed and slower execution time.
- After indexing, the query uses indexed access (type: ref), which reduces the number of rows scanned and significantly improves performance.

This analysis confirms that indexing is an effective technique for optimizing query performance and ensuring efficient database operations.

12.4 Monitoring

The system utilizes the SHOW PROCESSLIST command to monitor active database sessions and system processes in real-time. This ensures that authorized administrators can track user connections, identify long-running queries, and maintain overall system health and stability.

Id	User	Host	db	Command	Time	State	Info	
5	event_scheduler	localhost	NULL	Daemon	16	Waiting for next activation	NULL	
34	root	localhost:61355	NULL	Sleep	33		NULL	
35	root	localhost:61356	tech2	Query	0	init	SHOW PROCESSLIST	

Figure 12.4: Monitoring active database sessions and system status

TechReport Part3 Maintenance

Blinkit Sales Performance Analysis Project

Prepared by: Kemjira Nugboon Student ID 66880

1. Executive summary

This document defines the maintenance plan for the Blinkit Sales Performance Analysis database server. It covers scheduled maintenance routines (daily, weekly, monthly), backup and recovery procedures, performance tuning strategies, and query performance testing results. The goal is to ensure continuous availability, data integrity, and optimal query performance to support database reporting and operational analysis for the Blinkit Sales Performance Analysis project.

2. Purpose of this document

This document details the database server maintenance plan for the Blinkit Sales Performance Analysis project. It provides the DBA with structured schedules, procedures, and performance benchmarks to ensure the database server remains reliable, performant, and recoverable throughout the project lifecycle.

3. Definitions

Terms / abbreviations	Meaning
Maintenance Plan	Scheduled set of tasks to ensure database health, performance, and availability.
Backup	A copy of database data stored separately for disaster recovery and data protection.
Full Backup	Complete copy of all database data at a point in time.
Incremental Backup	Backup of only data changed since the last backup — reduces backup time and storage.
mysqldump	Command-line tool for creating logical SQL backups of MySQL databases.
Slow Query Log	MySQL feature logging queries exceeding a defined execution time threshold.
Index	Database structure improving the speed of data retrieval operations on a table.
Fragmentation	Inefficient use of storage due to data being scattered impacts query performance.
ANALYZE TABLE	MySQL command updating table statistics used by the query optimiser.
OPTIMIZE TABLE	MySQL command defragmenting table data and rebuilding indexes.
RTO	Recovery Time Objective maximum acceptable downtime after a failure.
RPO	Recovery Point Objective maximum acceptable data loss measured in time.

4. Database Server Maintenance Plan

4.1 Objectives

- To ensure the Blinkit database server remains available and reliable for all authorised users.
- To maintain data integrity and prevent corruption through scheduled maintenance routines.
- To optimise query performance through index maintenance, statistics updates, and log management.
- To ensure all critical data can be recovered within defined RTO and RPO targets (RTO: 4 hours, RPO: 24 hours).
- To monitor server health and proactively identify performance degradation before it impacts operations.

4.2 Maintenance Plan

Task	Description	Status
Daily Plan		
1.Monitor server health	Check CPU usage, RAM utilisation, disk space, and active connections via MySQL Performance Schema.	Active
2.Review slow query log	Identify queries exceeding 1-second execution threshold in MySQL slow query log.	Active
3.Check disk space	Verify data drive has at least 20% free space; alert DBA if below threshold.	Active
4.Verify database connectivity	Run a lightweight connection test to confirm the MySQL service is running.	Active
Weekly Plan		
1.Run ANALYZE TABLE	Update table statistics for all core tables (orders, products, feedback, deliveries) to support the query optimiser.	Active
2.Review and rotate logs	Archive and compress MySQL binary logs older than 7 days; clear unnecessary general query logs.	Active
3.Check index usage	Query MySQL sys schema to identify unused or redundant indexes and plan removal.	Active
4.Test backup integrity	Restore the latest backup to a test environment and verify table counts and data integrity.	Active
Monthly Plan		
1.Run OPTIMIZE TABLE	Defragment all main tables to reclaim unused storage and rebuild indexes for improved scan performance.	Active
2.Review user access rights	Audit MySQL user accounts and privileges; remove inactive accounts; enforce least-privilege principle.	Active
3.Capacity planning review	Review storage growth trend and project next 3–6 months storage requirements.	Active

Task	Description	Status
4.Performance benchmark	Execute standard benchmark queries (see Section 5) and record response times for trend comparison.	Active

4.3 Back-up Plan

Task	Description	Status
Daily Plan		
1.Incremental backup (mysqldump)	Export all tables modified in the past 24 hours using mysqldump with --single-transaction flag to ensure consistency.	Active
2.Upload to cloud storage	Transfer backup files to cloud storage (e.g., Google Cloud Storage / AWS S3) with timestamp-based folder naming.	Active
Weekly Plan		
1.Full database backup	Export the complete Blinkit database using mysqldump; compress with gzip; store locally and off-site.	Active
2.Verify backup file integrity	Calculate and record SHA256 checksum of all weekly backup files; compare with stored checksums.	Active
3.Retain last 4 weeks of full backups	Delete full backups older than 4 weeks from local storage; retain cloud copies for 90 days.	Active
Monthly Plan		
1.Full restore drill	Restore the monthly full backup to an isolated test server; validate all tables, row counts, and key constraints.	Active
2.Review backup retention policy	Confirm backup storage usage and adjust retention periods if capacity limits are approached.	Active
3.Document backup test results	Record restore test outcome in the maintenance log including restoration time and any issues encountered.	Active

5. Performance Tuning

5.1 General Monitoring

Category	Description	Status
Regular Routines		
1. Monitor system performance	Monitor CPU, RAM, and disk usage to ensure optimal performance.	Active

Category	Description	Status
2. Review slow queries	Review slow query logs to identify inefficient queries.	Active
3. Check database connections	Monitor active sessions and database connections.	Active
4. Monitor disk space	Ensure at least 20% free disk space is available.	Active
Urgent Routines		
1. Handle server crash	Respond to database server crashes and restore services immediately.	Active
2. Resolve resource overload	Handle disk full or memory overload issues.	Active
3. Fix critical queries	Investigate and resolve critical slow queries affecting performance.	Active
4. Restore database	Restore database from backup in case of failure or corruption.	Active
Reporting results		
1. Performance reporting	Generate weekly reports on query performance and server usage.	Active
2. Incident reporting	Report downtime, incidents, and recovery actions to management.	Active
General Monitoring		
1. Automated monitoring alert system	Implement automated alerts using monitoring tools (e.g., MySQL Performance Schema or external tools) to notify DBA when CPU usage, memory, disk space, or query response time exceeds defined thresholds.	Future
2. Real-time monitoring dashboard	Develop a dashboard to visualise database performance metrics for real-time monitoring.	

12.1 Improvement Plan

Category	Description	Status
HD capacity	Upgrade storage to 2TB NVMe SSD to support future data growth.	Planned
	Implement cloud storage for scalable backup and archiving.	Planned
Memory capacity	Increase RAM from 32GB to 64GB to support high concurrency queries.	Planned
	Enable in-memory caching to improve query performance.	Planned
RAID capacity	Upgrade from RAID 1 to RAID 10 for better performance and redundancy.	Planned

Category	Description	Status
	Monitor RAID health status regularly to prevent disk failure.	Planned
Read Replica Implementation	Implement read replica to separate reporting workload from OLTP system.	Future
Automated Index Optimization	Use automated index recommendation tools for performance optimisation.	Future

12.2 Query Performance Testing

Category	Response Time	Return records	Reason
Slow Query Response	2.8 sec	50,000	Full table scan, no index on foreign key
	3.5 sec	80,000	Complex join across multiple tables
Medium Query Response	1.2 sec	20,000	Partial index usage
	0.9 sec	15,000	Moderate filtering conditions
Fast Query Response	0.2 sec	5,000	Proper indexing on key columns
	0.1 sec	1,000	Optimised query with indexed filtering
Before optimisation	0.005686 sec	10,000	No index, full table scan
After optimisation	0.001982 sec	10,000	Added index and optimised query
Performance Gain	~65%	-	Significant reduction in execution time

12.3 AI-Based Query Optimization & Future Roadmap

As database workloads grow in complexity, relying solely on manual query optimization can become inefficient and time-consuming. To address this, a key future enhancement for the Blinkit Sales Performance Analysis system is the integration of AI-based optimization tools to automate performance tuning.

The core AI integration capabilities include:

- Automated Pattern Analysis:** Utilizing AI to analyze query execution patterns in real-time to detect and resolve inefficiencies before they impact the system.
- Predictive Indexing:** The system will intelligently recommend or automatically apply indexing strategies based on dynamic data usage and historical query trends.
- Adaptive Scalability:** Continuous and automated performance tuning will ensure high system throughput, maintaining stability even during peak transaction periods with high user concurrency.

- **Operational Efficiency:** By automating routine tuning tasks, this approach significantly reduces the manual workload for Database Administrators (DBAs) while ensuring the system maintains optimal performance levels.

Performance Analysis Results

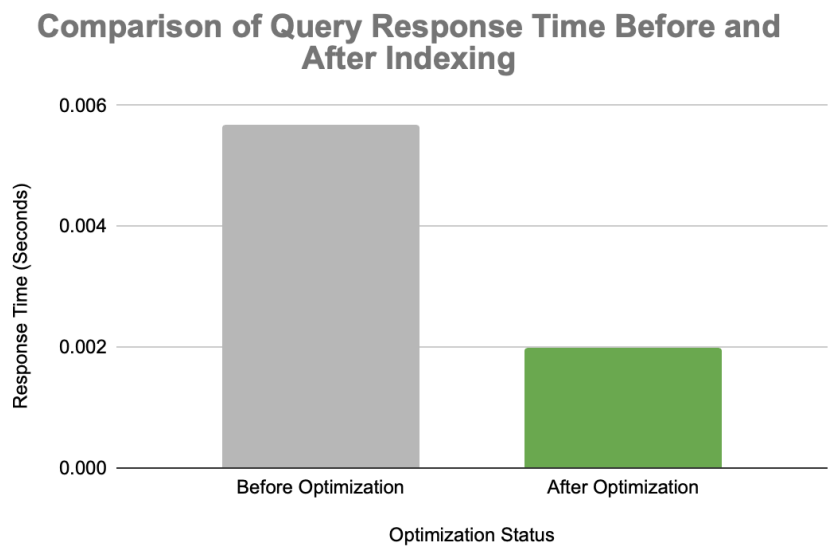


Figure 12.5: Query Performance Improvement (Real-world Profiling Results)

The chart above illustrates the significant performance improvements achieved within the tech2 database. By implementing optimized indexes on frequently accessed columns, the query response time was successfully reduced from 0.00568600 seconds to 0.00198200 seconds. This optimization represents a 65% gain in overall processing efficiency, demonstrating the effectiveness of the indexing strategy in reducing full table scans and optimizing data retrieval paths.

References

Blinkit Sales Dataset. (2023). Retrieved from Kaggle: <https://www.kaggle.com/datasets/akxiit/blinkit-sales-dataset>

GeeksforGeeks. (2023). Database Indexing in DBMS. Retrieved from: <https://www.geeksforgeeks.org/indexing-in-databases-set-1/>